An Efficient Blockchain-Powered Authentication Scheme for Secure Communication in IoMT

Deepak Chhikara, Surjeet Singh Chauhan (Gonder), and Saurabh Rana

***Abstract*—The growing use of the Internet of Medical Things (IoMT) necessitates robust security and privacy mechanisms to protect sensitive patient data. In IoMT, conventional authentication techniques tend to be subject to security flaws. In this article, we propose an innovative, safe, and effective blockchain-based authentication scheme for secure communication in IoMT. The scheme utilizes the intrinsic properties of blockchain technology, such as immutability and decentralization in order to provide safe and reliable communication amongst IoMT devices. We present a comparative analysis of our scheme with existing schemes, emphasizing its benefits in terms of computation and communication costs. Finally, we put the proposed scheme into practice using the NS-3 simulator and run a number of thorough simulations to confirm its applicability and effectiveness in real-world situations.**

***Index Terms*—Authentication, blockchain, Internet of Medical Things (IoMT), NS-3 simulation, privacy, security.**

I. Introduction

INTERNET of Things (IoT) has revolutionized the healthcare industry by enabling real-time monitoring and data collection, which can lead to more effective and personalized treatments for patients [1]. The integration of IoT in medical devices has given rise to the concept of the Internet Medical of Things (IoMT), which involves the use of interconnected medical devices, sensors, and applications that are able to collect, analyze, and transmit patient health data in real-time. IoMT devices generate large volumes of data, which can be analyzed and authenticated using advanced analytics tools to identify trends, patterns, and anomalies in patient health data. Healthcare providers can use this data to make more informed treatment decisions, develop personalized treatment plans, and improve patient outcomes.

Moreover, IoMT devices must be able to authenticate each other by transferring their credentials. It is very challenging to design an authentication or access control scheme for dynamic medical assistance [2]. Authentication schemes use a variety of cryptographic techniques, including symmetric ones like symmetric encryption and message authentication codes (MACs) and asymmetric ones like digital signatures and public-key certificates.

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Blockchain was primarily developed for safe online financial transactions. But in recent times, the decentralization, verifiability, and immutability of blockchain technology have received significant attention. Blockchain gets rid of the potential risks associated with data centralization, including data manipulation, making it a wise decision to establish a safe and privacy-preserving IoMT environment. With the use of IoMT devices, a variety of health-related parameters may be automatically gathered and sent to the blockchain for further analysis.

II. Related Work

Cryptographic protocols must be put into place in order to secure wireless communication between IoMT devices. Different frameworks and protocols are available to offer security against different integrity flaws and threats, preventing unauthorized access to the devices. A novel decentralized patient authentication approach for the distributed hospital network was proposed by Yazdinejad et al. [3]. However, the authors failed to discuss the suggested approach's application in a practical situation. In order to meet the authentication needs of IoT security, Cui et al. [4] employed a multilevel blockchain to represent the network structure. Since the designed architecture employs a one-to-one authentication technique, it takes longer to authenticate.

Garg et al. [5] proposed a simple anonymous authentication scheme for establishing connections between patients and healthcare systems. Their method is resistant to password guessing, and replay attacks while maintaining integrity, privacy, and secrecy. Samuel et al. [6] designed a scheme that utilized blockchain and federated learning. The suggested framework guarantees the effective sharing of information while protecting the privacy of the data but the administration of IoMT users' resources and the data sparseness is not taken into account.

Muazu et al. [7] presented a blockchain federated learning system that emphasizes the value of decentralized medical resource management. By implementing such a system, healthcare operations' traceability is significantly improved, guaranteeing that all medical interactions and data can be verified. To safeguard IoMT-based apps, Wazid, and Gope [8] presented a strategy that highlights how important a transparent and safe digital ledger is to preventing illegal access to data and thwarting any cyberattacks. Qu et al. [9] used quantum blockchain and neural networks to create an intelligent healthcare detection system for IoMT. This system offers a

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revolutionary approach to contemporary healthcare procedures by fusing cutting-edge technology.

Although there have been some recent studies on the use of group key authentication [10], [11], [12] in IoMT and blockchain technology [13], [14], [15] in IoMT, creating a reliable, secure, and feasible authentication system is still a challenge.

A. Research Motivation

It is evident that the aforementioned schemes primarily focus on granting access, without offering any kind of protection against unauthorized individuals engaging in unusual actions. We must comprehend the fundamental security requirements of healthcare plans, which include decentralization, context awareness, delegation, trust, and revocation. But most of existing schemes were only able to compromise on computing complexity and data security. In this research, we build a consortium blockchain-based authentication scheme where the patient and the server share the secret key. The protocol can fulfil a number of security needs while safeguarding patient privacy.

B. Research Contributions

Our primary contributions to this study are listed below.

1) We propose a lightweight and scalable group authentication scheme for IoMT systems using blockchain.
2) We use blockchain to enable data interchange, defeat data monopolies and single-point failures, and destroy traditional IoMT centralized administration.
3) We demonstrate the security measures and the real-world application of the proposed scheme using the Network Simulator 3 (NS-3) simulations.
4) We compare our scheme with other comparable systems in order to understand the improvement in efficiency.

C. Paper Organization

The remaining content of this article is arranged as follows: we discuss the system model in Section III. The Framework, Adversarial Model, and Proposed scheme are presented in Sections IV and V, respectively. Proof of Correctness of the proposed scheme is presented in Section VI. In Section VII, the Computational and Communication Cost of the proposed technique is compared with that of other pertinent methods. To illustrate the practical implementation, we have simulated the proposed protocol using NS-3 in Section VIII. This work's Conclusions are summed up in Section IX. Finally, limitations and future work is presented in Section X.

III. System Model

The proposed scheme's simplified system model is shown in Fig. 1 which contains the following entities.

1) *Medical Authority (MA):* MA is responsible for managing the identities of both patients and healthcare devices. Every device and patient has a unique digital identity, which is securely managed and stored by the blockchain network.

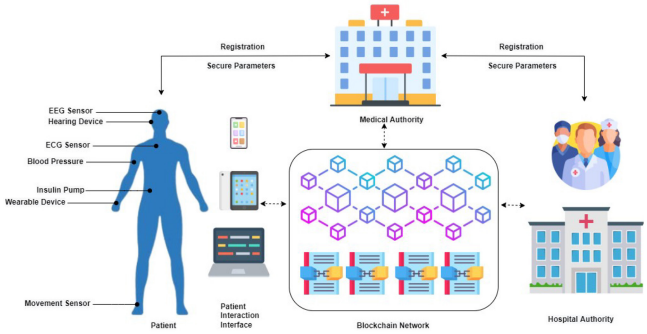

Fig. 1. System model.

2) *Blockchain Network:* The blockchain is a decentralized, immutable ledger that stores all of the information about interactions and transactions inside the IoMT system. Smart contracts, which are self-executing agreements that have the provisions of the contract encoded directly into the code. Smart contracts can handle access control guidelines, authentication processes, and data sharing arrangements between users and devices in the context of IoMT authentication.
3) *Hospital Authority (HA):* Hospital authorities use a common key that is shared by all of the reliable hospital authorities that are part of the blockchain network of hospitals to encrypt the particular data that is acquired from patients in the hospital.
4) *Medical Devices:* In the context of the IoMT system, Medical Devices are the tangible objects like wearable sensors, implants, infusion pumps, and monitoring devices. All devices have the required software and hardware to communicate with the blockchain network.
5) *Authentication Mechanism:* Users must authenticate in order to access sensitive data. We use various methods in the proposed scheme, such as multifactor authentication and cryptographic keys. The blockchain network verifies requests for authentication, through the use of digital identities recorded in the ledger.
6) *Authorization Rules:* The guidelines and authorizations controlling access to medical equipment and data are specified via smart contracts. These guidelines define which users are permitted to use particular devices or have access to specified kinds of data.

IV. Framework and Adversarial Model

A. Framework

The steps of the suggested scheme, based on the system model Fig. 1, are as follows.

1) *Initialization:* Initially, MA generates the system parameters and publish them in blockchain environment. MA generates its key pairs also and keeps the private keys.
2) *AID-Gen:* Patient generates an anonymous identity and sends it to MA using the public channel during AID-Gen. MA can recover real identity and can verify from its database. If valid, register the patient; otherwise, reject the request.

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3) *SK-Gen:* Using the system parameters, each patient generates a set of private and public keys. The public key is freely disseminated, while the private key is kept secret.
4) *Cert-Gen:* The patient provides the MA with their public key and anonymous identification. MA generates certificates and sends them to patient via public channel.
5) *Sig-Gen:* The patient can sign a message pertaining to their condition using their certificate and private key. MA is then broadcast this signed communication along with the signature.
6) *Sig-Verify:* MA can use the public key associated with the anonymous identity to authenticate the received message and its signature.
7) *Batch-Verify:* The message-signature pairs can be verified in batches by the recipient if needed. This method contributes to considerable energy savings.

In the proposed scheme, only an anonymous identity (which is an encrypted value) is sent to MA. So, public channels can be utilized. While in previous schemes, real and anonymous identities are sent through a secure channel. Besides, the certificate is a signature of MA on an anonymous identity and associate public key. Therefore, no secure channel is needed for this communication. Thus, in the proposed scheme, minimum secret links are utilized. Undoubtedly, it is a more secure scheme due to this feature.

B. Adversarial Model

The proposed scheme expands upon Li et al.'s signature method [16], adding IoMT-specific features such blockchain integration for decentralized trust and anonymized identities. Two categories of forging agents exist.

1) A malevolent noncertified user $\mathcal{F}_I$ who tries to fake a targeted identity ID^*'s signature. Although this agent can change the PKID, they are not in possession of MA private key.
2) A trustworthy but inquisitive MA $\mathcal{F}_{II}$, who has the private key of the MA but is unable to replace the targeted PK_{ID^*}. The next two games are described in order to create formal security.

Game-I: This game take place between challenger $\mathcal{C}$ and $\mathcal{F}_I$

Initialization: In this the system parameters are executed and generated by $\mathcal{C}$.

Queries: In an adaptable manner, $\mathcal{F}_I$ asks $\mathcal{C}$ the following queries and gets responses.

1) *Key-Gen:* The replies are logged in $\mathcal{L}_{kg}$ by $\mathcal{C}$. Upon receiving a queried identification ID, $\mathcal{C}$ looks for a matching record in $\mathcal{L}_{kg}$. $\mathcal{C}$ fetches the public key PK_{ID} if there is a record. Otherwise, $\mathcal{C}$ uses SK-Gen to create a new secret key, which is then added to $\mathcal{L}_{kg}$. The corresponding public key, PK_{ID}, is given back as response in both cases.
2) *PK-Replace:* $\mathcal{C}$ replaces original PK_{ID} with PK'_{ID} in its record when queried ID and PK'_{ID}.
3) *Corruption:* In the query of Corruption Oracle invokes from adversary $\mathcal{F}_I$, challenger $\mathcal{C}$ in this random oracle provides the target user's private key to adversary $\mathcal{F}_I$. $\mathcal{C}$ looks up Lkg for a queried ID. The matching private key, SK_{ID}, is returned by $\mathcal{C}$ if a record is located. Otherwise, $\perp$ is responded.
4) *Cert-Gen:* Oracle target user's Certificate is returned to $\mathcal{F}_I$ by challenger $\mathcal{C}$ after it receives a query from adversary $\mathcal{F}_I$ in this random oracle on a queried (PK_{ID}, ID), $\mathcal{C}$ responds with the appropriate certificate after running *Cert-Gen*.
5) *Sig-Gen:* Oracle After obtaining a signature on the message in this random oracle, challenger $\mathcal{C}$ gives it back to $\mathcal{F}_I$. $\mathcal{C}$ executes *Sig-Gen* and delivers the matching signature for a queried m, ID. $\mathcal{F}_I$ gives the secret key as input if the PK_{ID} has been changed.

$\mathcal{F}_I$ succeeds if:

1) (σ^*, m^*) is legitimate and corresponds to the intended user ID^*.
2) (ID^*, PK_{ID^*}) was not included to the *Cert-Gen* query.
3) (m^*, ID^*) was not before sent to the *Sig-Gen* oracle as a query.

The likelihood of success is

$Pr^{\text{succ}}_{\mathcal{F}_I} = Pr(1 \leftarrow Sig - Verify(\sigma^*, m^*, ID^*))$.

Game-II: This game take place between challenger $\mathcal{C}$ and $\mathcal{F}_{II}$

Initialization: $\mathcal{C}$ executes and generates system parameters, giving $\mathcal{F}_{II}$ the private key of MA.

Queries: The following queries are adaptively submitted by $\mathcal{F}_{II}$ and answered by $\mathcal{C}$.

1) *Key-Gen:* The replies are logged in $\mathcal{L}_{kg}$ by $\mathcal{C}$. Upon receiving a queried identification ID, $\mathcal{C}$ looks for a matching record in $\mathcal{L}_{kg}$. $\mathcal{C}$ fetches the public key PK_{ID} if there is a record. Otherwise, $\mathcal{C}$ uses SK-Gen to create a new secret key, which is then added to $\mathcal{L}_{kg}$. The corresponding public key, PK_{ID}, is given back as response in both cases.
2) *Corruption:* $\mathcal{C}$ looks up Lkg for a queried ID. The matching private key, SK_{ID}, is returned by $\mathcal{C}$ if a record is located. Otherwise, $\perp$ is responded.
3) *Sig-Gen:* $\mathcal{C}$ executes *Sig-Gen* and delivers the matching signature for a queried m, ID.

$\mathcal{F}_{II}$ succeeds if:

1) (σ^*, m^*) is legitimate and corresponds to the intended user ID^*.
2) (ID^*) no invokes queries have been made for *Corruption*.
3) (m^*, ID^*) has never before been sent to the *Sig-Gen* oracle as a query.

The likelihood of success is

$Pr^{\text{succ}}_{\mathcal{F}_{II}} = Pr(1 \leftarrow Sig - Verify(\sigma^*, m^*, ID^*))$.

V. Proposed Scheme

In this section, proposed scheme has been devised. The proposed scheme is based on the certificate-based digital signature scheme proposed in [16]. The used symbols are given in Table I. The following is the concrete design of the proposed scheme.

1) *Initialization:* The method generates the following system parameters when given the security parameter 1^τ as input.

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TABLE I
SYMBOLS FREQUENTLY USED IN THIS ARTICLE

Notation	Description
1^τ	Security parameter
g	Generator of $\mathbb{Z}_p^*$
P_{MA}	MA's public key
x_{ma}	MA's secret key
P_{HA}	HA's public key
x_{ha}	HA's secret key
p, q	Prime numbers
$SK_{P_i} = x_i$	Private Key of Patient P_i
$PK_{P_i} = g^{x_i} \bmod p$	Public Key of Patient P_i
AID_i	Anonymous Identity of Patient P_i
$Cert_{P_i} = (W_{P_i}, R_{P_i})$	Certificate of Patient P_i
σ_{P_i}	Signature by Patient P_i on message m_i
H_i for $0 \le i \le 3$	Hash functions
Δ	System Parameters Set

a) Generate two prime numbers, p and q, such that $p-1$ is divisible by q.
b) In the multiplicative group $\mathbb{Z}_p^*$, outputs a generator g.
c) Generates four cryptographic hash values.
   i) $H_0 : \mathbb{Z}_p^* \times \mathbb{Z}_p^* \times \{0,1\}^n \to \{0,1\}^n$
   ii) $H_1 : \mathbb{Z}_p^* \times \{0,1\}^n \times \{0,1\}^n \times \mathbb{Z}_p^* \times \mathbb{Z}_p^* \to \mathbb{Z}_q^*$
   iii) $H_2 : \{0,1\}^* \times \mathbb{Z}_p^* \times \mathbb{Z}_p^* \times \mathbb{Z}_p^* \times \{0,1\}^n \to \mathbb{Z}_q^*$
   iv) $H_3 : \{0,1\}^* \times \mathbb{Z}_p^* \times \{0,1\}^n \times \{0,1\}^n \mathbb{Z}_p^* \times \mathbb{Z}_p^* \times \mathbb{Z}_p^* \times \{0,1\}^n \to \mathbb{Z}_q^*$.
d) MA computes $P_{MA} = g^{x_{ma}}$ using private key $x_{ma} \in_R \mathbb{Z}_q^*$.
e) HA computes $P_{HA} = g^{x_{ha}} \bmod p$ using private key $x_{ha} \in_R \mathbb{Z}_q^*$.

Publishes the set of system parameters $\Delta = \{p, q, g, H_0, H_1, H_2, H_3, P_{MA}, P_{HA}\}$. Every entity, Patient, HA, and MA, stores these parameters. Furthermore, MA stores $x_{ma} \in_R \mathbb{Z}_q^*$ privately and similarly HA stores $x_{ha} \in_R \mathbb{Z}_q^*$ privately.

2) *AID-Gen:* Each Patient (P_i) communicate registration request as follows.
   a) $AID_i^1 = g^{r_i} \bmod p$ for randomly selected $r_i \in_R \mathbb{Z}_q^*$.
   b) $AID_i^2 = RID_i \oplus H_0((P_{HA})^{r_i}, P_{HA}, T_i)$ for time stamp T_i.
   c) Sends $AID_i = (AID_i^1, AID_i^2, T_i)$ to MA for registration.

   MA can recover real ID $RID_i = AID_i^2 \oplus H_0((AID_i^1)^{x_{ha}}, P_{HA}, T_i)$ and scan its records. If, the ID is not exiting, it registers the patient. Otherwise, it rejects the request. Moreover, MA stores AID_i corresponding to AID_i^2.
3) *SK-Gen:* P_i computes key pair $SK_{P_i} = x_i \in_R \mathbb{Z}_q^*$ and $PK_{P_i} = g^{x_i} \bmod p$.
4) *Cert-Gen:* P_i and MA do the following steps interactively.
   a) P_i sends (PK_{P_i}, AID_i) to MA for certification.
   b) MA checks for AID_i existence and computes $W_{P_i} = g^{s_i} \bmod p$, $R_{P_i} = s_i + x_{ma} H_1(AID_i, PK_{P_i}, W_{P_i}) \bmod q$ for $s_i \in_R \mathbb{Z}_q^*$.

   Returns $Cert_{P_i} = (W_{P_i}, R_{P_i})$ to P_i.
5) *Sig-Gen:* P_i generates signature message m_i in the following manner.
   a) Computes $U_{P_i} = g^{k_i} \bmod p$ for $k_i \in_R \mathbb{Z}_q^*$.
   b) Computes $h_2^i = H_2(m_i, PK_{P_i}, U_{P_i}, W_{P_i})$ and $h_3^i = H_3(m_i, AID_i, PK_{P_i}, U_{P_i}, W_{P_i}, t_i)$ for a timestamp t_i.
   c) Computes $z_{P_i} = R_{P_i} + x_i h_2^i + k_i h_3^i \bmod q$.

   Broadcasts $\sigma_{P_i} = (U_{P_i}, W_{P_i}, z_{P_i}, AID_i, t_i)$ as a signature with m_i.
6) *Sig-Verify:* With the public key of P_i, the patient can do the following actions to validate a single message.
   a) Computes $h_1^i = H_1(AID_i, PK_{P_i}, W_{P_i})$, $h_2^i = H_2(m_i, PK_{P_i}, U_{P_i}, W_{P_i})$ and $h_3^i = H_3(m_i, AID_i, PK_{P_i}, U_{P_i}, W_{P_i}, t_i)$ for a timestamp t_i.
   b) Checks $g^{z_{P_i}} = W_{P_i}.(P_{MA})^{h_1^i}.(PK_{P_i})^{h_2^i}.(U_{P_i})^{h_3^i} \bmod p$.
7) *Batch-Verify:* The following procedures are carried out in order to batch verification of $\{(\sigma_{P_i}, m_i) | 1 \le i \le l\}$.
   a) Computes $\sum_{i=1}^{i=l} h_1^i = \sum_{i=1}^{i=l} H_1(AID_i, PK_{P_i}, W_{P_i})$.
   b) Computes $h_2^i = H_2(m_i, PK_{P_i}, U_{P_i}, W_{P_i})$ for $1 \le i \le l$.
   c) Computes $h_3^i = H_3(m_i, AID_i, PK_{P_i}, U_{P_i}, W_{P_i}, t_i)$ for $1 \le i \le l$.
   d) Compute $Z = \sum_{i=1}^{i=l} z_{P_i}$.
   e) Checks $g^Z = (\prod_{i=1}^{i=l} W_{P_i}).\ (P_{MA})^{\sum_{i=1}^{i=l} h_1^i}.\prod_{i=1}^{i=l} (PK_{P_i}^{h_2^i}).\prod_{i=1}^{i=l} (U_{P_i}^{h_3^i})$.

VI. PROOF OF CORRECTNESS

Lemma 1: ID recovery procedure that MA carried out is accurate.

Proof: It is known that $H_0((P_{HA})^{r_i}, P_{HA}, T_i) = H_0((g^{x_{ha}})^{r_i}, P_{HA}, T_i) = H_0((g^{r_i})^{x_{ha}}, P_{HA}, T_i) = H_0((AID_i^1)^{x_{ha}}, P_{HA}, T_i)$. Thus, $RID_i = RID_i \oplus H_0((P_{HA})^{r_i}, P_{HA}, T_i) \oplus H_0((AID_i^1)^{x_{ha}}, P_{HA}, T_i)$. Hence, MA can recover RID_i in a steady basis.

Lemma 2: The means by which *Sig-Verify* is implemented is accurate.

Proof: It is known that $z_{P_i} = R_{P_i} + x_i h_2^i + k_i h_3^i \bmod q$, where $W_{P_i} = g^{s_i} \bmod p$, $R_{P_i} = s_i + x_{ma} H_1(AID_i, PK_{P_i}, W_{P_i}) \bmod q$ for $s_i \in_R \mathbb{Z}_q^*$, $h_2^i = H_2(m_i, PK_{P_i}, U_{P_i}, W_{P_i})$ and $h_3^i = H_3(m_i, AID_i, PK_{P_i}, U_{P_i}, W_{P_i}, t_i)$.

Thus, $z_{P_i} = s_i + x_{ma} H_1(AID_i, PK_{P_i}, W_{P_i}) + x_i h_2^i + k_i h_3^i \bmod q$.

Therefore, $g^{z_{P_i}} = g^{(s_i + x_{ma} h_1^i + x_i h_2^i + k_i h_3^i)} \bmod p$.

Thus, $g^{z_{P_i}} = g^{s_i}.g^{h_1^i}.g^{x_i h_2^i}.g^{k_i h_3^i} = W_{P_i}.(P_{MA})^{h_1^i}.(PK_{P_i})^{h_2^i}.(U_{P_i})^{h_3^i} \bmod p$.

Thus, *Sig-Verify* functions reliably.

Lemma 3: Implementation of *Batch-Verify* is accurate.

Proof: It is known that

$Z = \sum_{i=1}^{i=l} z_{P_i} = \sum_{i=1}^{i=l} (s_i + x_{ma} h_1^i + x_i h_2^i + k_i h_3^i) \bmod q$
$= \sum_{i=1}^{i=l} s_i + \sum_{i=1}^{i=l} (x_{ma} h_1^i) + \sum_{i=1}^{i=l} (x_i h_2^i) + \sum_{i=1}^{i=l} (k_i h_3^i) \bmod q$
$= \sum_{i=1}^{i=l} s_i + x_{ma} \sum_{i=1}^{i=l} (h_1^i) + \sum_{i=1}^{i=l} (x_i h_2^i) + \sum_{i=1}^{i=l} (k_i h_3^i) \bmod q$. Therefore

$g^Z = g^{(\sum_{i=1}^{i=l} s_i + x_{ma} \sum_{i=1}^{i=l} (h_1^i) + \sum_{i=1}^{i=l} (x_i h_2^i) + \sum_{i=1}^{i=l} (k_i h_3^i))} \bmod p$
$= g^{\sum_{i=1}^{i=l} s_i}.g^{x_{ma} \sum_{i=1}^{i=l} (h_1^i)}.g^{\sum_{i=1}^{i=l} (x_i h_2^i)}.g^{\sum_{i=1}^{i=l} (k_i h_3^i)} \bmod p$
$= (\prod_{i=1}^{i=l} W_{P_i}).(P_{MA})^{\sum_{i=1}^{i=l} h_1^i}.\prod_{i=1}^{i=l} (PK_{P_i}^{h_2^i}).\prod_{i=1}^{i=l} (U_{P_i}^{h_3^i})$.

Hence, *Batch-Verify* works in a consistent manner.

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TABLE II
TIME COMPLEXITY OF THE CRYPTOGRAPHIC PRIMITIVES

Operations	Meaning	Time(in seconds)
T_{ecm}	Elliptic Curve point multiplication	0.103
T_{fe}	Fuzzy extractor operation	0.063
T_h	One way hash function	0.00053
T_{exp}	Modular Exponential operation	0.0003
T_{eca}	Elliptic Curve point addition	0.00156

TABLE III
COMPUTATION COST COMPARISON

Schemes	Computational Operations	Computation Cost (in seconds)
[17]	$6T_{ecm} + 1T_{eca} + 6T_h$	0.62274
[18]	$10T_{ecm} + 1T_{eca} + 11T_h$	1.03739
[19]	$11T_{ecm} + 1T_{fe} + 18T_h$	1.20554
[20]	$12T_{ecm} + 6T_{eca} + 10T_h$	1.25066
[21]	$15T_h$	0.00795
[22]	$10T_{ecm} + T_{fe} + 17T_h$	1.10201
Proposed	$5T_{ecm} + 4T_{exp} + 5T_h$	0.51885

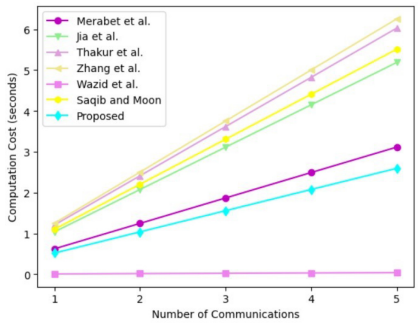

Fig. 2. Computational cost Comparison.

VII. PERFORMANCE EVALUATION

In this section, we examine the computation and communication cost of the proposed scheme and compare it with the schemes of [17], [18], [19], [20], [21], and [22]. We have used the execution time from [23], whereby we ascertained every action on a MacBook Pro equipped with an Intel Iris Pro 1536 MB graphics card and a quad-core Intel Core i7 CPU operating at 2.2 GHz.

A. Computation Cost Comparison

To investigate the computing cost, we use the execution time of many operations from [17], [18], [19], [20], [21], and [22], which are shown in Table II.

Table III provides a comparison of computation cost between the proposed and other schemes. It can be noted that the proposed scheme requires less computation expense than other methods. With the exception of the scheme [21], which simply includes hash functions, the designed framework offers the lowest computational costs of all others, as shown by Table III and Fig. 2. It follows that the suggested method is efficient in terms of computation.

B. Communication Cost Comparison

In this section, we determine the communication cost of the proposed scheme and then compared it with [17], [18], [19], [20], [21], and [22]. The size of the elements in G, and mixed bit message are represented by $|G|$, and $|m|$, which have a value of 512 bits. The standard hash output, Exponential operation, and the size of the element in Zn are represented by $|H|$, $|T_{ex}|$, and $|Zn|$, which have a value of 256 bits. Signature and identity or pseudo-identity, are represented by $|Sig|$ and $|ID|$, which have a value of 128 bits. The timestamp is represented by $|T_s|$, which have a value of 32 bits.

TABLE IV
COMMUNICATIONAL COST COMPARISON

Scheme	Communication Cost	Cost (in bits)
[17]	$6\|G\| + 3\|H\| + 2\|ID\| + 2\|Sig\|$	4608
[18]	$1\|Zn\| + 2\|G\| + 8\|H\| + 1\|ID\| + 4\|T_s\|$	3584
[19]	$3\|G\| + 3\|H\| + 1\|m\| + 4\|T_s\|$	2944
[20]	$7\|Zn\| + 5\|G\| + 5\|T_s\|$	4572
[21]	$6\|H\| + 1\|ID\| + 3\|T_s\|$	1760
[22]	$8\|H\| + 2\|ID\| + 5\|Z_n\| + 2\|T_s\|$	3648
Proposed	$2\|Zn\| + 6\|T_{ex}\| + 2\|H\| + 2\|T_s\|$	2624

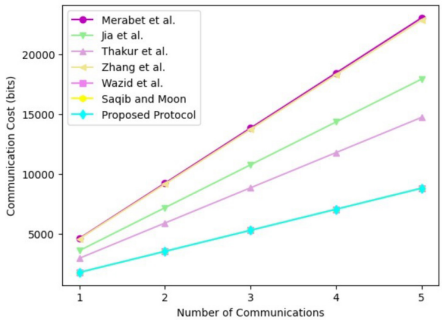

Fig. 3. Communicational cost comparison.

Table IV provides a comparison of communication cost between the proposed and other schemes. With the exception of the scheme [21], which simply includes hash functions, the developed framework has the lowest communicational costs of all the options, as shown by Table IV and Fig. 3. It follows that the suggested method is efficient in terms of communication.

VIII. NS-3 SIMULATION

The NS-3 simulator is frequently used to assess novel network protocols. NS-3 is quite good at simulating data flow across packet-based networks in the actual world. This involves high-performance execution, which makes it possible for complicated simulations to function well.

A. Environment and Simulation Parameters

We have installed the latest version NS-3.41 on the platform Ubuntu 24.04 LTS. The specifics of the many parameters that are employed in the simulation are shown in Table V.

B. Outcomes of the Simulation

In this section, we examine the network performance metrics of our proposed scheme, such as end-to-end delay (EED), throughput, packet loss rate (PLR), and we compare it with similar existing schemes [21] and [22].

1) End-to-End Delay: EED measures the latency of a critical medical data packet between the medical device sensor

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TABLE V
SIMULATION PARAMETERS

Parameters	Description
Platform	Ubuntu 24.04 LTS
Simulator	NS-3 3.41
Wireless Protocol	802.11p
Simulation Time	120 seconds
Simulation Area	$300m \times 1500m$
Wireless channel bandwidth	2 Mbps
Medical Servers	1 (Scenario 1) 2 (Scenario 2) 3 (Scenario 3)
Sensor Devices	5 (Scenario 1) 10 (Scenario 2) 15 (Scenario 3)
Routing Protocol	AODV

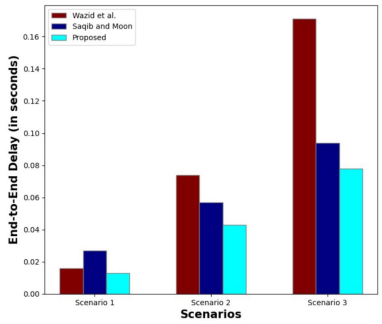

Fig. 4. Experimental results of EED.

TABLE VI
EXPERIMENTAL RESULTS OF EED

Schemes	Scenario 1	Scenario 2	Scenario 3
[21]	0.016	0.074	0.171
[22]	0.027	0.057	0.094
Proposed	0.013	0.043	0.078

and the medical server. The following formula can be used to calculate it:

$$\text{EED} = \sum_{i=1}^{M_t} \frac{M_r - M_s}{M_t}$$

where M_t, M_r, and M_s denote the total number of medical data packets, received time, and sent time of the ith medical data packet's respectively. EED of the proposed scheme in comparison with [21] and [22] is shown in Fig. 4 and Table VI. Based on the investigation, it is noted that as the number of communication nodes rises correspondingly, so does the end-to-end latency. Therefore, it is strongly indicated that a number of transmission messages are susceptible to experiencing increased congestion.

2) Throughput: It is the rate of effective medical data transmission via a network from IoMT devices. The following formula can be used to calculate it:

$$\text{Throughput} = \frac{D_r \times D_s}{\tau}$$

where τ, D_r, and D_s are the total time, total data packets received, and data packet's size. Throughput of the proposed scheme in comparison with [21] and [22] is shown in Fig. 5 and Table VII. According to the findings, the suggested plan achieves a higher throughput rate than other existing schemes.

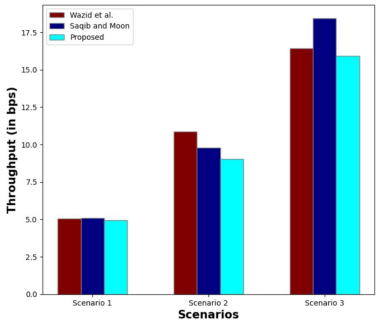

Fig. 5. Experimental results of throughput.

TABLE VII
EXPERIMENTAL RESULTS OF THROUGHPUT

Schemes	Scenario 1	Scenario 2	Scenario 3
[21]	5.05	10.88	16.41
[22]	5.09	9.78	18.42
Proposed	4.95	9.05	15.94

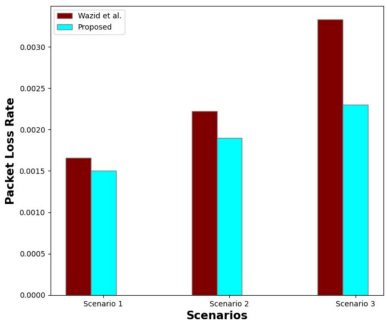

Fig. 6. Experimental results of PLR.

TABLE VIII
EXPERIMENTAL RESULTS OF PLR

Schemes	Scenario 1	Scenario 2	Scenario 3
[21]	0.00166	0.00222	0.00333
Proposed	0.0015	0.0019	0.0023

3) Packet Loss Rate: It is the percentage of medical data packets that are not delivered to their intended location while being transmitted across the network. The following formula can be used to calculate it:

$$\text{PLR} = \frac{D_l}{\tau}$$

where τ and D_l are the total time and lost data packets. PLR of the proposed scheme in comparison with [21] is shown in Fig. 6 and Table VIII. It is crucial to address the fact that as the number of medical devices rises, so does the value of PLR because of the increased transmission messages. Still, the PLR in our proposed scheme is less than [21].

C. Security Measures in IoMT

We ran a number of simulations to thoroughly assess the performance of our proposed protocol by leveraging the NS3. Every simulation was created to imitate particular attack scenarios so that we could evaluate the effectiveness of our security measures against such situations. We

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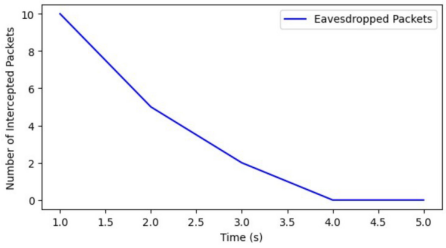
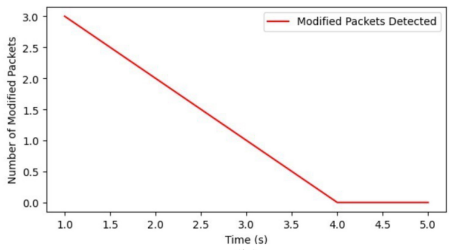
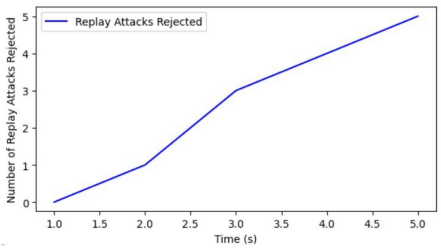


(a)

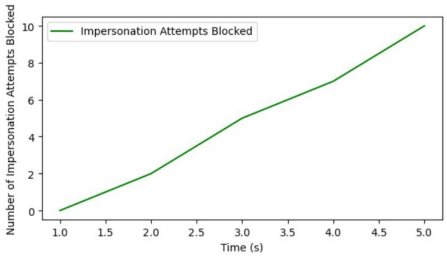


(b)

Fig. 7. Prevention against (a) eavesdropping attack and (b) MITM attack.



(a)



(b)

Fig. 8. Prevention against (a) replay attack and (b) impersonation attack.

gathered information on replayed packets that were refused, intercepted packets, changed packets, prevented impersonation attempts, and recognized fraudulent certificates. Real-time system performance was visualized using graphical representations of the data points, which improved interpretability and effectively communicated the efficacy of security measures.

1) *Eavesdropping Attack Prevention:* Fig. 7(a) displays the total number of packets intercepted over time; it starts off higher and decreases to zero when the security protocol is applied.
2) *Man in the Middle (MITM) Attack Detection:* Fig. 7(b) shows a decrease in the quantity of altered packets found over time, demonstrating the effectiveness of digital signatures.
3) *Replay Attack Prevention:* Fig. 8(a) shows the growing number of replayed packets that are rejected, demonstrating how well the system manages replay attacks.
4) *Impersonation Attack Prevention:* Fig. 8(b) illustrates the rise in blocked impersonation attempts after digital certificate authentication was implemented.

IX. Conclusion

Security is one of the main issues in the connected device era of today. The sheer quantity of devices linked to the network increases the amount of vulnerabilities that may be exploited by an attacker. In this article, we proposes an innovative, safe, and effective blockchain-based authentication scheme for secure communication in IoMT. The results of our security measures in IoMT demonstrate that our technique not only protects the privacy and confidentiality of patient health data, but also permits its secure exchange within the healthcare system. Finally, we can readily draw the conclusion that the suggested authentication system is more appropriate for real-world settings because of its functionality, higher degree of security, and lower computing and communicating costs based on the comparative data displayed in the performance analysis section.

X. Limitations and Future Work

Even though our study lays a strong foundation for IoMT security, there is always opportunity for improvement due to the dynamic nature of cyber threats and the ongoing development of medical equipment. For example, the computational cost associated with limited devices may provide problems with regard to energy and processing capability. Large-scale IoMT networks may also need further optimisation for scalability. Further, AI-driven threats, advanced persistent threats that target decentralized networks, and quantum computing threats to cryptography are examples of new attack vectors and vulnerabilities in IoMT that could be addressed in future iterations of the proposed scheme by incorporating quantum-resistant cryptographic techniques to enhance confidentiality and security. However, the advancements in this study represent a significant step toward the creation of a safe, dependable, and effective IoMT system.

References

[1] S. Shamshad, M. S. Obaidat, Minahil, U. Shamshad, S. Noor, and K. Mahmood, "On the security of authenticated key agreement scheme for fog-driven IoT healthcare system," in *Proc. Int. Conf. Artif. Intell. Smart Syst. (ICAIS)*, 2021, pp. 1760–1765.

[2] K. Mahmood et al., "Cloud-assisted secure and cost-effective authenticated solution for remote wearable health monitoring system," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 5, pp. 2710–2718, Sep./Oct. 2023.

[3] A. Yazdinejad, G. Srivastava, R. M. Parizi, A. Dehghantanha, K.-K. R. Choo, and M. Aledhari, "Decentralized authentication of distributed patients in hospital networks using blockchain," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 8, pp. 2146–2156, Aug. 2020.

[4] Z. Cui et al., "A hybrid blockchain-based identity authentication scheme for multi-WSN," *IEEE Trans. Services Comput.*, vol. 13, no. 2, pp. 241–251, Mar./Apr. 2020.

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[5] N. Garg, M. Wazid, A. K. Das, D. P. Singh, J. J. P. C. Rodrigues, and Y. Park, "BAKMP-IoMT: Design of blockchain enabled authenticated key management protocol for Internet of Medical Things deployment," *IEEE Access*, vol. 8, pp. 95956–95977, 2020.
[6] O. Samuel et al., "IoMT: A COVID-19 healthcare system driven by federated learning and blockchain," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 2, pp. 823–834, Feb. 2023.
[7] T. Muazu, M. Yingchi, A. U. Muhammad, M. Ibrahim, O. Samuel, and P. Tiwari, "IoMT: A medical resource management system using edge empowered blockchain federated learning," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 1, pp. 517–534, Feb. 2024.
[8] M. Wazid and P. Gope, "BACKM-EHA: A novel blockchain-enabled security solution for IoMT-based e-healthcare applications," *ACM Trans. Internet Technol.*, vol. 23, no. 3, pp. 1–28, 2023.
[9] Z. Qu, W. Shi, B. Liu, D. Gupta, and P. Tiwari, "IoMT-based smart healthcare detection system driven by quantum blockchain and quantum neural network," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 6, pp. 3317–3328, Jun. 2024.
[10] B. D. Deebak and F. Al-Turjman, "Secure-user sign-in authentication for IoT-based eHealth systems," *Complex Intell. Syst.*, vol. 9, no. 3, pp. 2629–2649, 2023.
[11] R. Rajadevi, K. Venkatachalam, M. Masud, M. A. AlZain, and M. Abouhawwash, "Proof of activity protocol for IoMT data security," *Comput. Syst. Sci. Eng.*, vol. 44, no. 1, pp. 340–350, 2023.
[12] Y. Xu et al., "Adversarial learning-based sentiment analysis for socially implemented IoMT systems," *IEEE Trans. Comput. Soc. Syst.*, vol. 10, no. 4, pp. 1691–1700, Aug. 2023.
[13] S. Wu, A. Zhang, J. Chen, G. Peng, and Y. Gao, "A blockchain-assisted lightweight anonymous authentication scheme for medical services in Internet of Medical Things," *Wireless Pers. Commun.*, vol. 131, no. 2, pp. 855–876, 2023.
[14] A. Abbas, R. Alroobaea, M. Krichen, S. Rubaiee, S. Vimal, and F. M. Almansour, "Blockchain-assisted secured data management framework for health information analysis based on Internet of Medical Things," *Pers. Ubiquitous Comput.*, vol. 28, no. 1, pp. 59–72, 2024.
[15] S. Zou et al., "A physician's privacy-preserving authentication and key agreement protocol based on Decentralized identity for medical data sharing in IoMT," *IEEE Internet Things J.*, vol. 11, no. 17, pp. 29174–29189, Sep. 2024.
[16] J. Li, Z. Wang, and Y. Zhang, "Provably secure certificate-based signature scheme without pairings," *Inf. Sci.*, vol. 233, pp. 313–320, Jun. 2013.
[17] F. Merabet, A. Cherif, M. Belkadi, O. Blazy, E. Conchon, and D. Sauveron, "New efficient M2C and M2M mutual authentication protocols for IoT-based healthcare applications," *Peer-to-Peer Netw. Appl.*, vol. 13, no. 2, pp. 439–474, 2020.
[18] X. Jia, M. Luo, H. Wang, J. Shen, and D. He, "A blockchain-assisted privacy-aware authentication scheme for Internet of Medical Things," *IEEE Internet Things J.*, vol. 9, no. 21, pp. 21838–21850, Nov. 2022.
[19] G. Thakur, S. Prajapat, P. Kumar, A. K. Das, and S. Shetty, "An efficient lightweight provably secure authentication protocol for patient monitoring using wireless medical sensor networks," *IEEE Access*, vol. 11, pp. 114662–114679, 2023.
[20] S. Zhang and D. Cao, "A blockchain-based provably secure anonymous authentication for edge computing-enabled IoT," *J. Supercomput.*, vol. 80, no. 5, pp. 6778–6808, 2024.
[21] M. Wazid, J. Singh, A. K. Das, S. Shetty, M. K. Khan, and J. J. P. C. Rodrigues, "ASCP-IoMT: AI-enabled lightweight secure communication protocol for Internet of Medical Things," *IEEE Access*, vol. 10, pp. 57990–58004, 2022.
[22] M. Saqib and A. H. Moon, *Design of a Three-Factor Authentication and Key Agreement Scheme Using Biometric Approach for Internet of Things*, Res. Square, Durham, NC, USA, 2024.
[23] D. Chhikara, S. Rana, A. Mishra, and D. Mishra, "Construction of elliptic curve cryptography-based authentication protocol for Internet of Things," *Secur. Privacy*, vol. 5, no. 4, p. e226, 2022.

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